ANALYTICS CAPSTONE · OCTOBER 2026

Executive Decision Report

Where growth is coming from, where it leaks, and three decisions worth making this quarter.

PREPARED FOR
Executive Leadership Team

BASIS
Customer EDA (5,000 members) and KPI dashboard (24 months, 16 states)

HORIZON
12-month plan, Nov 2026 to Oct 2027



EXECUTIVE SUMMARY

Growth is strong, but retention and acquisition efficiency leave money on the table

REVENUE, LAST 12 MONTHS
$2.37M
+26.5% vs prior 12 months

CHURN RATE
4.00%
-1.00 pp, improving

CUSTOMER ACQUISITION COST
$27.18
-1.5% vs prior, but Paid Ads costs $46

WHAT WE FOUND

- **Volume drives growth.** About 80% of the revenue increase is more orders; order value added 4.9%.
- **Deep discounts backfire.** Customers at 30%+ discount retain at 36% against 62% overall.
- **Service failures predict churn.** Customers with 3+ tickets score 6.6 satisfaction vs 7.9, and retain at 48%.
- **Paid Ads is 53% of spend** at 6.2% churn, 2.6x Email.

WHAT WE RECOMMEND

1. **Cap standard discounts at 20%** and win back the 30%+ cohort. ROI +135%.
2. **Pilot a ticket early-warning programme** with a stage gate. ROI -11% base, +56% upside.
3. **Shift 15% of Paid Ads volume** to organic and email. ROI +98%.

Base case: $42.8K 12-month benefit on $23.4K of programme cost, ROI +83%, payback in about 6.6 months. Conservative case is roughly break-even (-1%).

DECISION CONTEXT

Leadership needs to decide where the next 12 months of retention and acquisition effort go

THREE QUESTIONS THIS REPORT ANSWERS

1. **What is driving revenue?**
   Which levers, segments and categories explain the growth?

2. **Where is value leaking?**
   Which customers and channels lose money or churn fastest?

3. **What should we do, and what will it return?**
   Which actions, in what order, with what payback?

WHAT WAS ANALYSED

Scope A: customer base

`5,000 customers · $4.4M annual spend · 61.7% retained`

Descriptive statistics, correlations and three hypothesis tests on discount rate, segment and support load.

Scope B: commerce performance

`24 months · 16 states · 5 categories · 4 channels`

Revenue, AOV, CAC and churn from the executive KPI dashboard, with year-on-year comparison.

METHODOLOGY

Findings rest on tested relationships, with effect sizes reported next to p-values

Prepare

Quality checks, median imputation of 120 missing satisfaction and 60 missing age values (flagged), duplicate check.

Describe

Mean, median, standard deviation, quartiles and skew. Outliers via the 1.5x IQR and z-score rules.

Test

Chi-square (discount vs retention), Kruskal-Wallis (AOV by segment), Mann-Whitney (satisfaction by tickets). Alpha 0.05, plus Cramer's V, epsilon-squared, Cohen's d.

Model

Spearman correlations and a logistic regression to separate the effects of satisfaction, tenure, discount and tickets on retention.

Quantify

Three-scenario ROI model per recommendation, with every assumption listed on the ROI slide.

TOOLS AND DELIVERABLES

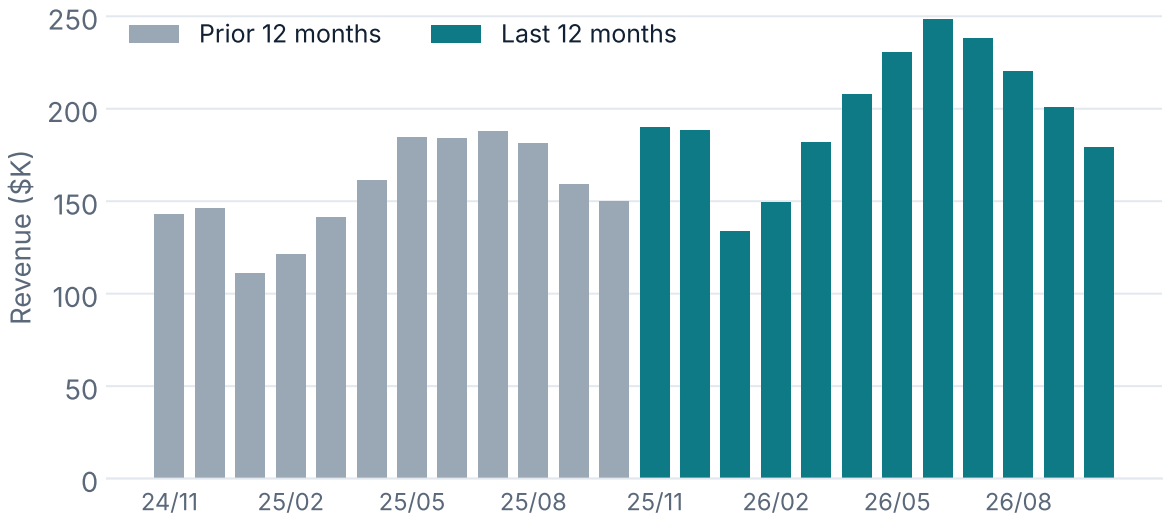
Python (pandas, SciPy, statsmodels, seaborn) in a Jupyter notebook; Streamlit and Plotly dashboard with a PDF export; this report.

READ THIS BEFORE RELYING ON THE NUMBERS

Data is synthetic and seeded for reproducibility. Patterns show the method; dollar figures are scenario estimates, not forecasts. Associations are not proof of cause, so each recommendation carries a stage gate.

PERFORMANCE SNAPSHOT

Revenue grew 26.5% year on year, mostly from order volume



ORDERS
25,058
+20.6%

AVG ORDER VALUE
$94.57
+4.9%

CAC
$27.18
-1.5%

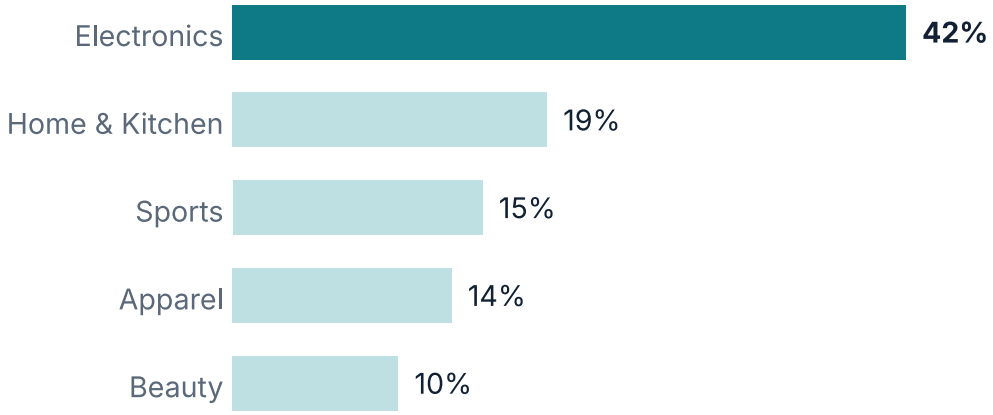
CHURN
4.00%
-1.00 pp

Revenue of $2.37M is +26.5% on the prior 12 months. Orders account for about 80% of the growth, so the business is scaling through volume, not price.

KEY REVENUE DRIVERS

Order value and category mix drive revenue; Electronics alone is 42%

SHARE OF REVENUE BY CATEGORY

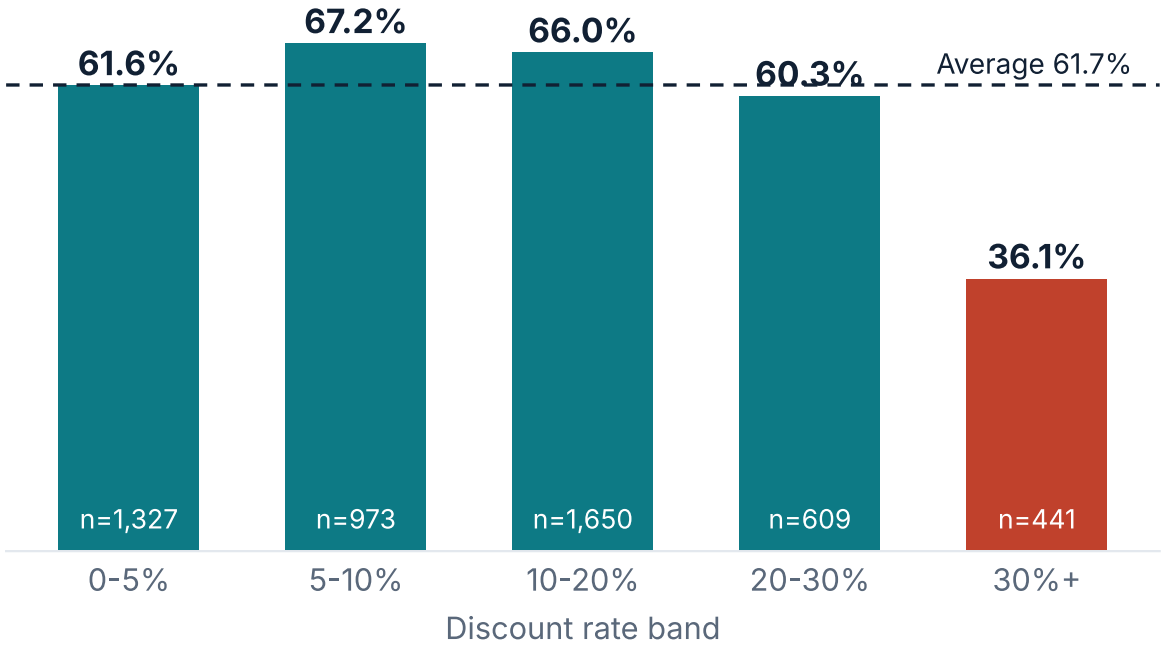


Commerce data. Electronics averages $193 per order against $48 for Beauty. By region: South 30%, Northeast 27%, West 23%, Midwest 20%.

WHAT MOVES REVENUE (CUSTOMER FILE)

- **Basket size beats frequency.** Customer spend correlates 0.83 with order value but only 0.48 with orders per year (Spearman).
- **Segments differ sharply.** Median order value is $50 Consumer, $109 Small Business, $308 Enterprise (Kruskal-Wallis p near 0, epsilon-squared 0.67).
- **Retention does not vary by segment** (62%, 61%, 63%), so segment is a revenue lever, not a churn lever.

Discounts help up to about 20%; beyond 30%, retention falls to 36%



DEEP-DISCOUNT COHORT

441

8.8% of customers

AVERAGE DISCOUNT

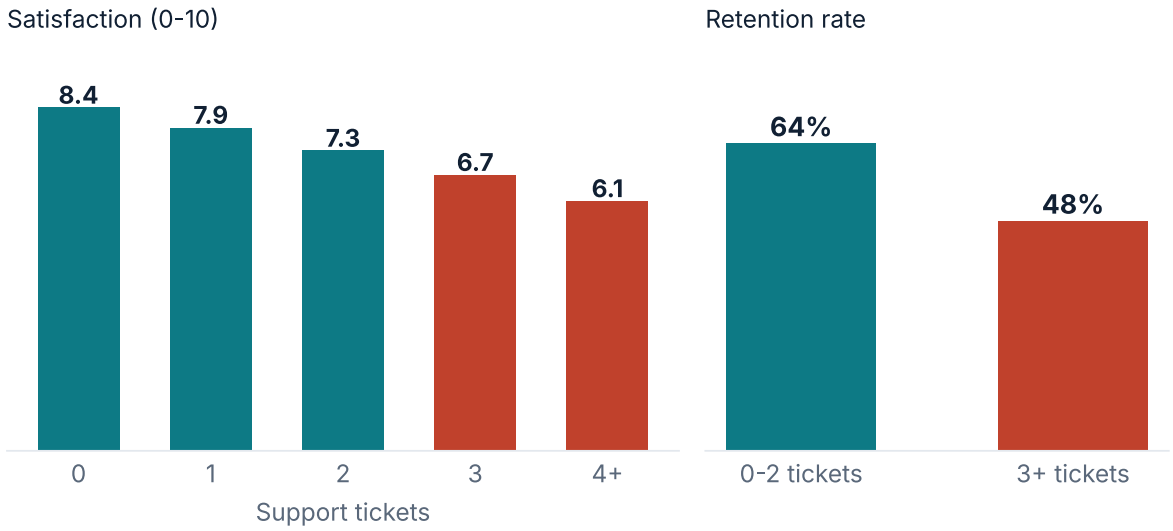
36%

in that cohort

- Differences across bands are significant: chi-square 148.8, p < 1e-30, Cramer's V 0.17.
- A straight-line view hides it: the linear correlation is only -0.07. The effect is a curve, peaking at 5-20%.
- The logistic model agrees: a positive discount term (odds ratio 1.91) and a strong negative squared term (0.38).

RETENTION DRIVER 2: SERVICE

Three or more support tickets is the clearest early warning of churn



AT-RISK CUSTOMERS

627

12.5% of base

SATISFACTION GAP

1.3 pts

6.6 vs 7.9

- Mann-Whitney p < 1e-120, Cohen's d -1.16: a very large effect.
- Each standard deviation of satisfaction raises the odds of staying by 63% (odds ratio 1.63).
- Retention is 48% for 3+ tickets against 64% for the rest.

RISK AREAS

Five risks could erode growth; three are addressed directly by the recommendations

RISK	EVIDENCE	SEVERITY	RESPONSE
Deep-discount churn	441 customers at 36% average discount retain at 36%	High	Rec 1
Service-driven churn	627 customers with 3+ tickets retain at 48%; satisfaction 6.6	Medium	Rec 2
Paid Ads dependency	53% of spend, CAC $46, churn 6.2% vs 2.4% Email	High	Rec 3
Revenue concentration	Electronics 42% of revenue; CA, NY, TX together 34%	Medium	Monitor
Data quality	2.4% satisfaction scores missing; 504 order-value outliers by IQR rule (10%), skew of 4.1	Low	Validate before forecasting

Severity reflects the size of the affected group and the gap in retention or cost against the average. It is a judgement for prioritisation, not a measured score.

RECOMMENDATION 1

Cap standard discounts at 20% and win back the 30%+ cohort

WHAT WE DO

- Set 20% as the ceiling for standard offers; route anything higher through an approval rule.
- Replace deep discounts for the 441 affected customers with targeted win-back offers and service touchpoints.
- Track retention of the cohort monthly against the 20-30% band benchmark of 60%.

Why: this cohort retains 24.2% points below the 20-30% band, and the effect is statistically solid ($p < 1e-30$).

BASE CASE, 12 MONTHS

INVESTMENT
$11.3K
$6,000 set-up + $12 per customer

BENEFIT
$26.5K
75 extra retained customers

ROI
+135%
+34% conservative, +236% upside

BREAK-EVEN
30%
of the retention gap closed

Assumes 70% of the gap closes; each retained customer contributes $355 ($1,015 spend x 35% margin). Margin saved from lower discounts is not counted.

RECOMMENDATION 2

Pilot a ticket early-warning programme, and scale it only if it clears the break-even gate

WHAT WE DO

- Flag any customer reaching a third ticket and trigger proactive outreach from a senior agent within 48 hours.
- Offer a service-recovery gesture where the ticket was caused by us.
- Run for 90 days on the 627 at-risk customers; hold back a control group so the effect is measured, not assumed.

Stage gate: expand only if the pilot closes at least 22% of the 16-point retention gap. Below that, stop and fix the root causes of tickets instead.

BASE CASE, 12 MONTHS

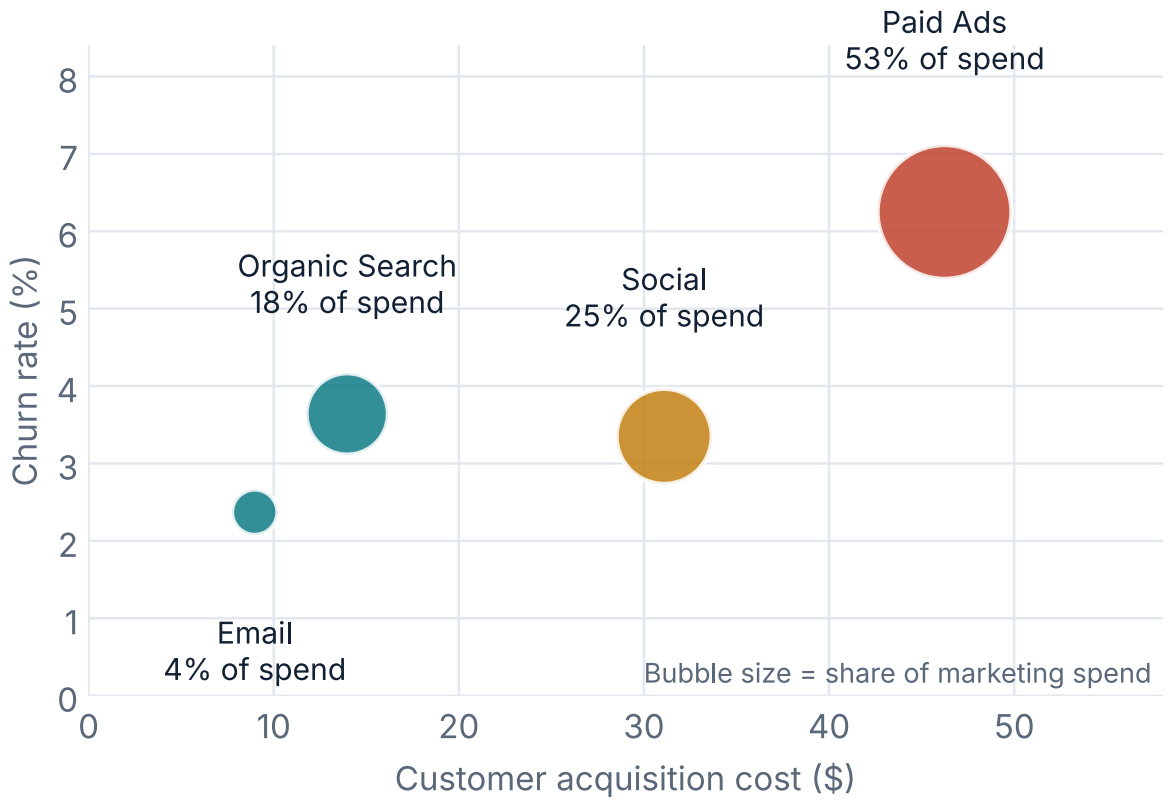
INVESTMENT	BENEFIT
$7.1K	**$6.3K**
$4,000 tooling + $5 per customer	20 extra retained customers

ROI	BREAK-EVEN
-11%	**22%**
-56% conservative, +56% upside	of the retention gap closed

Assumes 20% of the gap closes (much of the gap may not be caused by tickets). Retained value is $315 per customer. Multi-year value and lower cost-to-serve are not counted.

RECOMMENDATION 3

Shift 15% of Paid Ads volume to organic and email acquisition



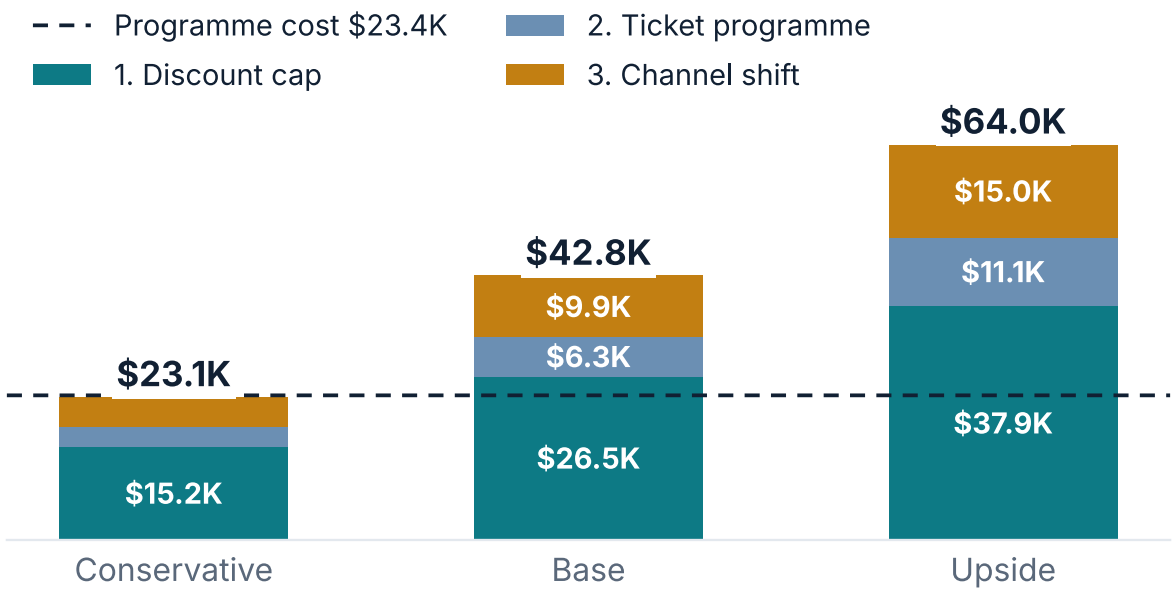
- Paid Ads takes 53% of marketing spend and has the highest CAC ($46.24) and churn (6.2%).
- Organic Search costs $13.97, Email $8.98; both churn less.
- Moving 15% of Paid Ads volume (391 customers) at 1.5x today's organic CAC ($20.96) saves $25.28 per customer.

INVESTMENT	BENEFIT	ROI
$5.0K	**$9.9K**	**+98%**
content, CRM, analytics	acquisition cost saved	-5% to +200%

The 1.5x penalty allows for diminishing returns. Lower churn of the moved customers is not counted, so this is conservative.

ROI PROJECTIONS

Combined, the three actions return +83% in the base case and break even in the worst case



	CONSERV.	BASE	UPSIDE
12-month benefit	$23.1K	$42.8K	$64.0K
Programme cost	$23.4K	$23.4K	$23.4K
Net benefit	-$0.3K	$19.4K	$40.6K
ROI	-1%	+83%	+173%
Payback (months)	12.2	6.6	4.4

KEY ASSUMPTIONS

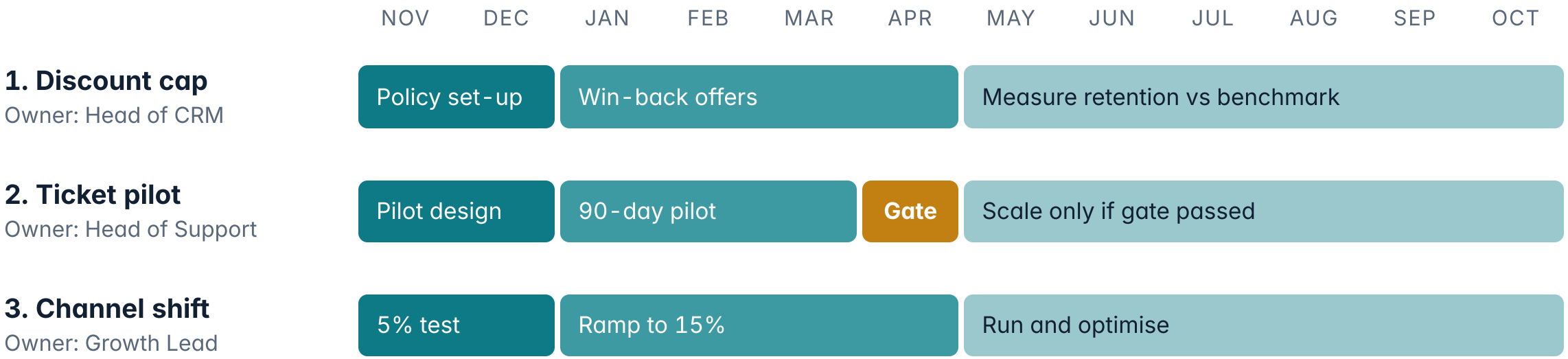
35% contribution margin; 12-month value only. Retention gap closed (conservative/base/upside): Rec 1 40%/70%/100%, Rec 2 10%/20%/35%. Costs are planning estimates.

NOT MODELLED

Margin saved on lower discounts, lower churn of channel-shifted customers, multi-year value, cost-to-serve savings. Figures are for a base of 5,000 customers and scale with it.

IMPLEMENTATION TIMELINE

A 12-month plan with one explicit stage gate and three checkpoints



KPIS TO TRACK

Retention of 30%+ cohort · retention and satisfaction of 3+ ticket customers · blended CAC and Paid Ads share of spend · overall churn.

CHECKPOINTS

Jan: first retention read on the cohort. **Apr:** ticket-pilot gate. **Jun:** mid-year ROI review against this report.

MAIN DELIVERY RISKS

Sales pushback on the discount cap; support capacity for outreach; organic volume that does not scale at 1.5x CAC.

Three approvals today start the plan; the first results arrive in January

DECISIONS REQUESTED

1. **Approve the 20% discount cap** and the win-back programme for the 441-customer cohort ($11.3K).
2. **Approve the 90-day ticket pilot** with a control group and a 22% break-even gate ($7.1K).
3. **Approve moving 15% of Paid Ads volume** to organic and email in two stages ($5.0K).

Total ask: $23.4K. Owners named on the previous slide.

NEXT 30 DAYS

- Finance replaces planning costs and margin with actuals and re-runs the ROI model.
- CRM builds the cohort list and offer rules; Support defines the third-ticket trigger.
- Growth sets the 5% test budget and success thresholds.

SUPPORTING PORTFOLIO

Final analytical notebook (EDA, hypothesis tests, findings), interactive KPI dashboard with PDF export, and the code to reproduce every number in this report.